

Answer Key — General Science Grade 10

Sub-Strand 1.3: Nutrition in Animals • Lesson 4: Food Tests Practical Part 1 — Detecting Starch and Reducing Sugars in Kenyan Foods

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
A	D	A	C	D	C	D

USE AFTER: PREDICT PHASE — Predict: prediction card with Think-Pair-Share on ugali and ripe banana

Q1. Before testing ugali, a learner recalls the enzymes from the last lesson. Which food substance does salivary amylase act on?

- A. Starch**
- B. Water
- C. Fat
- D. Protein

Correct answer: A. Salivary amylase digests starch, beginning in the mouth, and produces maltose. Protein and fat are digested by other enzymes.

USE AFTER: OBSERVE PHASE — Observe: iodine test and Benedict test on the five food samples, including the glucose solution control

Q2. A group tests a glucose solution as the control. What results should they expect?

- A. Iodine: blue-black; Benedict: stays blue
- B. Iodine: brown-orange; Benedict: stays blue
- C. Iodine: blue-black; Benedict: orange-red after heating
- D. Iodine: brown-orange; Benedict: orange-red after heating**

Correct answer: D. Glucose is a reducing sugar, not starch. Iodine stays brown-orange (no starch) and Benedict reagent turns orange-red when heated in a hot water bath.

USE AFTER: OBSERVE PHASE — Observe: iodine test comparing cooked ugali and raw maize flour

Q3. Raw maize flour turns blue-black when two drops of iodine are added. What does this result show?

- A. Starch is present in the maize flour even before it is chewed**
- B. Starch is absent from the maize flour
- C. Reducing sugars are present in the maize flour
- D. Salivary amylase has already digested the starch

Correct answer: A. Blue-black with iodine is a positive test for starch. Digestion by salivary amylase only starts when the food is chewed and mixed with saliva, so starch is still present in raw maize.

USE AFTER: OBSERVE PHASE — Observe: teacher demonstration of Benedict test safety and heating in the hot water bath

Q4. During the Benedict test, a test tube is heated in a hot water bath for five minutes. What is the safe way to handle the tube?

- A. Leave the tube in the boiling water bath and lift it out with your hand when the colour changes
- B. Hold it with bare fingers so you can feel when it is hot enough
- C. Use a test-tube holder and keep the tube at an angle, pointing away from your face and others**
- D. Hold it upright with a test-tube holder and look down into it

Correct answer: C. The tube is hot and its contents can spurt out. Use a holder and point the tube away from yourself and others.

USE AFTER: EXPLAIN PHASE — Explain: group paragraph, prompt 2 (why ripe banana gives a positive Benedict result)

Q5. A ripe banana gives an orange-red result with Benedict reagent, but an unripe banana gives a much weaker result. What best explains this?

- A. Benedict reagent detects starch, which is present in ripe bananas
- B. Ripe bananas never contained starch at any stage
- C. Iodine solution has changed the banana into sugar
- D. During ripening, starch in the banana is converted into reducing sugars**

Correct answer: D. As a banana ripens, its starch is changed to reducing sugars, so the Benedict test becomes positive and the banana tastes sweeter. Benedict reagent detects reducing sugars, not starch.

USE AFTER: EXPLAIN PHASE — Explain: board evidence statement 'EVIDENCE FROM FOOD TEST — UGALI CONTAINS STARCH'

Q6. Cooked ugali gives a blue-black colour with iodine. Which is the best evidence statement to write in your notebook?

- A. Ugali contains no carbohydrate, because it does not taste sweet
- B. Ugali contains starch, which is the substrate for lipase
- C. Ugali contains starch, which is a substrate for salivary amylase and pancreatic amylase**
- D. Ugali contains reducing sugars, because starch and sugar are the same substance

Correct answer: C. Blue-black with iodine shows starch. Starch is digested by salivary amylase in the mouth and pancreatic amylase in the small intestine. Starch and reducing sugar are different substances, and lipase acts on fats.

USE AFTER: MODEL BUILDING PHASE — Model: two-box diagram (MOUTH and SMALL INTESTINE) using the ugali example

Q7. A learner draws a two-box diagram for ugali: Box 1 is MOUTH and Box 2 is SMALL INTESTINE. Which pair of labels is correct?

- A. Mouth: no digestion of starch happens; small intestine: salivary amylase digests all the starch
- B. Mouth: salivary amylase breaks starch into maltose; small intestine: lipase digests the remaining starch
- C. Mouth: pepsin breaks starch into maltose; small intestine: salivary amylase digests the remaining starch
- D. Mouth: salivary amylase starts breaking starch into maltose; small intestine: pancreatic amylase digests the remaining starch**

Correct answer: D. Starch digestion starts in the mouth with salivary amylase, which produces maltose. Any remaining starch is digested by pancreatic amylase in the small intestine. Lipase acts on fats, not starch.